Generalizable Deep Learning for Aerosol and Cloud Segmentation in Ground-Based and Spaceborne Lidar Observations

Adrián Canella-Ortiz, Siham Tabik, Sol Fernández-Carvelo, Onel Rodríguez-Navarro, Lucas Alados-Arboledas, and Ana del Águila

***Abstract*—Automated interpretation of lidar observations remains a major challenge in atmospheric remote sensing due to the complexity, variability, and large volume of acquired measurements. Existing approaches for aerosol and cloud identification typically depend on handcrafted algorithms and expert-defined thresholds, limiting their robustness and scalability across instruments and observation conditions. This work presents a deep learning (DL) framework for the semantic segmentation of atmospheric structures directly from multiwavelength lidar signals. The proposed methodology represents time-height range-corrected signal measurements as images and exploits computer vision segmentation architectures to identify aerosols and clouds. To enable the training and rigorous evaluation of DL segmentation models, a new expert-annotated dataset, named DL4Lidar, was constructed from one year of observations acquired with a ground-based Raman lidar system. Extensive experiments were conducted to analyze the impact of lidar wavelength, signal representation, preprocessing strategies, backbone architectures, multiscale feature extraction, and boundary refinement modules. The resulting models achieved Intersection over Union values of 0.814 for aerosols and 0.753 for clouds on unseen test data, demonstrating accurate segmentation under diverse atmospheric scenarios. The framework also showed strong cross-domain robustness when applied, without retraining, to measurements from an independent ground-based lidar instrument, obtaining comparable performance. In addition, the methodology was successfully extrapolated to satellite-based atmospheric lidar observations from the EarthCARE mission, enabling accurate segmentation of cloud structures and retrieval of cloud top height temporal evolution. These results demonstrate that semantic segmentation provides a powerful and transferable paradigm for automated analysis of both ground-based and spaceborne lidar observations, opening new opportunities for scalable atmospheric monitoring and exploitation of large lidar archives. The associated dataset (Zenodo: https://doi.org/10.5281/zenodo.20156234) and code (GitHub: https://github.com/acanort/DL4Lidar.git) have been made publicly available.**

***Index Terms*—Aerosol, cloud, computer vision, deep learning (DL), generalization, lidar, satellite, semantic segmentation.**

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I. Introduction

ACCURATE identification of aerosols and clouds in the atmosphere is essential for improving the predictive capabilities of Earth system models and understanding the climate processes. While clouds strongly influence the Earth's radiative balance, aerosols significantly modulate cloud formation and precipitation mechanisms, thereby affecting both weather and climate dynamics [1]. Their misrepresentation has long been recognized as one of the dominant sources of uncertainty in climate simulations, as highlighted by all previous Intergovernmental Panel on Climate Change (IPCC) assessment reports [2]. Reliable detection and classification of aerosols and clouds, therefore, remain key prerequisites for reducing uncertainties in climate feedbacks and radiative forcing estimates [3].

Ground-based active remote sensing instruments, such as multiwavelength lidars, provide high-resolution vertical profiles of atmospheric backscatter signals, enabling the retrieval of intensive optical properties, such as extinction coefficients, depolarization ratios, and Ångström exponents, that are crucial for characterizing atmospheric components. Long-term networks, including aerosol, clouds and trace gases research infrastructure (ACTRIS) [4], European aerosol research lidar network (EARLINET) [5], E-Profile [6], Latin American Lidar network (LALINET) [7], and micro-pulse

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lidar network (MPLNET) [8], have established standardized measurement protocols that have yielded extensive archives of lidar observations and supported a large body of [9], [10], [11]. Despite their high scientific impact, a substantial fraction of these archives remains challenging to exploit in a fully automated and scalable manner, as the classification of atmospheric features (e.g., cloud layers, aerosol plumes, or clear-sky periods) often relies on complex post-processing algorithms or rule-based thresholding schemes that require significant expert intervention [12], [13], [14]. Moreover, these traditional approaches frequently employ computationally intensive processing pipelines that scale poorly to large observational archives, limiting their suitability for real-time atmospheric monitoring and large-scale climatological studies.

Beyond ground-based systems, spaceborne lidars, such as the atmospheric Lidar (ATLID) instrument on board the EarthCARE mission [15] also provide vertically resolved backscatter measurements at a global scale, enabling synergistic studies between ground-based and satellite platforms. Developing robust, automated classification frameworks for lidar data is therefore essential not only for the exploitation of extensive ground-based archives but also for the future integration and analysis of satellite-based observations (e.g., [16], [17], [18], [19], [20], [21]).

The rapid advancement of machine learning (ML) has created new opportunities for automating the analysis of ATLID observations. ML methods can learn underlying patterns directly from the data, enabling large-scale, efficient, and potentially more objective analysis of atmospheric signals. Several studies have explored the use of ML for lidar-based applications, including aerosol typing [22], [23], [24], [21], and aerosol layering [25]. However, most existing approaches focus on derived products rather than on the original lidar signal, which limits their ability to represent the spatial and temporal variability of atmospheric structures.

In parallel, deep learning (DL)-based segmentation methods have demonstrated remarkable performance in satellite remote sensing applications involving cloud detection and atmospheric feature identification. For instance, Li et al. [26] proposed the MSCFF model, which uses an encoder-decoder architecture to detect clouds and cloud shadows across a wide range of Earth observation sensors, while Jeppesen et al. [27] introduced RS-Net, a U-Net-based network for cloud segmentation in Landsat-8 imagery, demonstrating robust performance even under challenging surface conditions. More recently, DL methods have also been applied to aerosol-cloud classification in satellite lidar observations using U-Net-based architectures trained on operational retrieval products [28], [29]. These studies highlight the potential of convolutional-based segmentation models to capture spatial and contextual patterns relevant for atmospheric component classification. However, such approaches remain largely unexplored for ground-based lidar observations, which offer complementary vertical and temporal information to satellite measurements.

A promising alternative emerges from treating lidar signals as images, thus framing the problem as one of semantic segmentation, that is, a pixel-wise classification task in which each pixel of an input image is assigned a label corresponding to a predefined category. By converting time-height sections of the range-corrected signal (RCS) into image representations, atmospheric classification can benefit from well-established computer vision segmentation models that have revolutionized domains, such as medical imaging [30] and autonomous driving [31].

In this work, we propose a DL framework for the semantic segmentation of aerosols and clouds using image representations of the multiwavelength lidar RCS. Unlike conventional ML approaches that depend on derived optical properties, our method directly processes the lidar backscatter signal, allowing the model to learn spatial structures and patterns inherent in the atmospheric profiles. To this end, we constructed a new human expert-annotated dataset (DL4Lidar) from the ALHAMBRA lidar system operated at the ACTRIS/EARLINET Granada station [32]. Expert-labeled datasets are extremely scarce in the ATLID community, yet they are crucial for training and validating supervised models. The dataset was specifically designed to enable the supervised training and validation of segmentation models under real observational conditions.

The pipeline proposed in this work, including data preprocessing, annotation protocol, and model design, is conceived to be generalizable to other lidar systems and applicable across atmospheric monitoring networks. Furthermore, the proposed methodology is readily transferable to emerging satellite-based lidars, such as ATLID, on board the EarthCARE satellite [15], offering automated aerosol and cloud detection across both ground-based and spaceborne platforms. To assess its scalability beyond ground-based configurations, we implemented a preliminary application to the ATLID instrument lidar observations, which demonstrated consistent performance and strong potential for future operational use. Therefore, the methodology developed in this work demonstrates that a model trained solely on ground-based data can accurately reproduce cloud structures observed from space, highlighting the robustness and transferability of the learned morphological representations with semantic segmentation.

II. Methodology

In this work, we address the problem of cloud and aerosol identification in lidar measurements as a semantic segmentation problem. Methodologically, our goal is to learn structural and morphological delineations from expert-validated annotations on ground-based RCS data. To develop a model capable of identifying these atmospheric components in lidar-based images, we followed three main steps outlined in Fig. 1.

1) Construction of DL4Lidar, a high-quality dataset for training and evaluating the segmentation model, consisting of images generated from RCS data over continuous 5-h intervals, and manually annotated for aerosol and cloud classification.
2) Identification of the best segmentation strategy, data representation, and data preprocessing that increase the discrimination capacity between the cloud and aerosol classes during the learning process, performed through an iterative process.

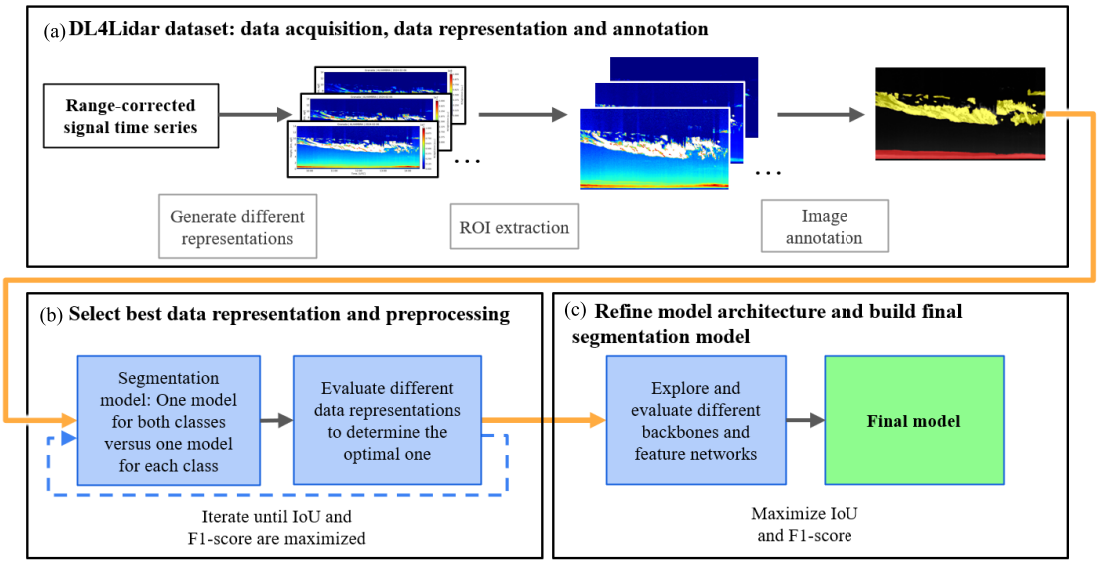


Fig. 1. Flowchart of the proposed methodology. (a) Construction of the DL4Lidar dataset, where the ROI, i.e., the portion of the image where the lidar signal is plotted, is extracted and annotated. (b) Determination of the segmentation model baseline and optimal data representation. (c) Enhancement of the designed segmentation model by finding the best-performing architecture.

3) Improvement of the robustness and generalization capacity of the final model by analyzing different feature extractor architectures, also known as backbones, and feature-network variants. To ensure the reliability of the evaluation and to avoid overfitting, a fivefold cross-validation scheme was applied throughout steps (2) and (3).

Sections II-A–II-C provide details on the construction of the DL4Lidar dataset and the design of the segmentation model.

A. DL4Lidar Dataset

1) Data Acquisition: The construction of the DL4Lidar dataset is based on measurements acquired from September 2023 to August 2024 by a ground-based $3\beta + 3\alpha + 2\delta$ Dual-LMRD multispectral lidar system known as ALHAMBRA [32], [33], manufactured by Raymetrics S.L. Here, β, α, and δ denote the aerosol backscatter coefficient, extinction coefficient, and depolarization, respectively. The system is operated and located at the University of Granada (UGR), Spain, as part of the Andalusian Global Observatory of the Atmosphere (AGORA), integrated within EARLINET [5] and the European research infrastructure ACTRIS [4].

ALHAMBRA is a dual multiwavelength Raman lidar designed for high-resolution vertical profiling of the atmosphere. It combines two optical subsystems designed to cover both the atmospheric boundary layer and the lower stratosphere, near-field and far-field, respectively, and provides elastic backscatter signals at 355, 532, and 1064 nm, along with vibrational and rotational Raman, depolarization, and fluorescence measurements. Data are acquired at a temporal resolution of 1 min and a vertical resolution of 3.75 m, offering fine-scale information on atmospheric structures.

We have considered using the RCS due to its suitability for conversion into visual formats compatible with DL workflows. The RCS represents the lidar signal received from the backscattering of emitted laser pulses by atmospheric constituents, corrected for the decrease in signal intensity with distance by multiplying it by the square of the range [34]. Before computing the RCS, the raw signal measured by the instrument undergoes standard preprocessing procedures commonly applied in ATLID measurements [35], [36], including bin-zero correction, dead-time correction, background noise removal, and dark-current correction.

These preprocessing steps are required to obtain physically consistent and instrument-corrected lidar signals before further analysis. In particular, additive noise contributions, such as background radiation and electronic noise, must be removed prior to RCS computation, since the range correction amplifies residual noise at large distances, potentially introducing artificial high-intensity structures and degrading segmentation reliability. In addition, to preserve the fine structures relevant for downstream ML tasks, no smoothing or normalization is performed.

Due to the high operational costs associated with running this lidar system, measurements are performed mainly on demand, generally following the requirements of CAMS [37] and EarthCARE. These measurement sessions typically last from 4 to 6 h. Therefore, the data are segmented into fixed 5-h intervals. Each 5-h segment is transformed into an image representing the temporal evolution of the RCS, where the vertical axis corresponds to height (0-14 km above ground level, agl), the horizontal axis to time, and the color scale encodes the RCS values expressed in arbitrary units (a.u.), allowing the visualization of aerosol layers and clouds. An example of this representation is shown in Fig. 2.

Note that when the 5-h intervals have periods with missing data, gray vertical stripes are used to fill those gaps. This is a common situation, as measurements may be interrupted for several reasons, such as low clouds, rainfall over the instrument, or other operational issues.

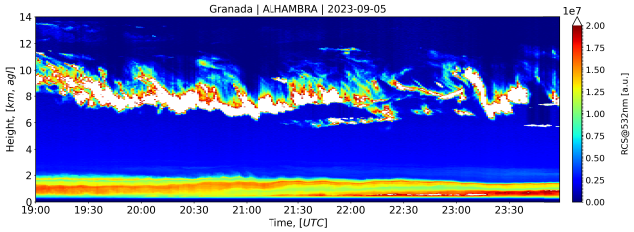


Fig. 2. Representation of RCS measured on 2023-09-05 during nighttime at 532 nm over a 5-h period.

2) Data Representation: To address the aerosol-cloud identification problem using DL-based computer vision methods, the lidar RCS measurements were transformed into image representations. For this purpose, we used the GFATpy Python package [38], which processes all available measurement channels of the ALHAMBRA lidar and provides flexibility in the visual output by allowing adjustments to parameters, such as saturation limits, color palettes, maximum display height, selected time range, and the variable to be plotted. These configurations enabled the generation of different image representations, which were subsequently used as input to the segmentation models.

Exploring such representations is consistent with findings in visual perception research, as several influential studies on understanding the differences between machine and human visual perception have shown that modern vision models can exploit spectral intervals broader than those accessible to humans [39], [40]. Inspired by these studies, we evaluated the impact of different data representations by considering different combinations of measured data at different wavelengths. In particular, among a large number of possible combinations, we empirically found that the combinations with high potential for improving the performance of the segmentation model are.

1) Use of RCS data measured at different fixed wavelengths. The impact of each elastic channel, namely, 355, 532, and 1064 nm, is analyzed individually. Fixed color scales were used to plot the data.
2) Averaging RCS data measured at different wavelengths using a common maximum scale. Since aerosols and clouds exhibit enhanced backscatter signals at different wavelengths, combining the elastic signals from the three channels can emphasize features that remain undetected in single-wavelength RCS images.
3) Averaging RCS data at the same wavelength with different maximum scales. Using a similar argument to the previous preprocessing, the choice of the maximum scale influences the saturation and contrast of the image, potentially enhancing the visibility of relevant atmospheric features.

By systematically generating and testing these alternative image inputs, we assess how different wavelengths and scale bounds influence the segmentation performance for aerosol and clouds.

The generated images contain nonrelevant elements, such as axis ticks, titles, and legends (see Fig. 2). We remove these elements using an automated cropping-based algorithm. Only the region containing actual RCS data is preserved and used as model input, as can be seen in Fig. 1, after the "Region of Interest (ROI) Extraction" step. If the images are generated directly from the RCS data and contain only the data region, this cropping step can be omitted.

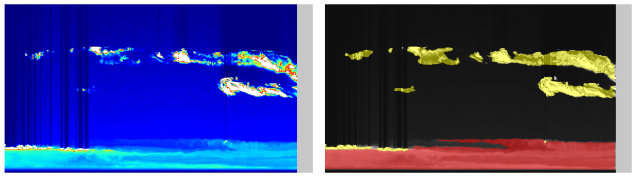


Fig. 3. Example of one image annotation. The left panel shows the ROI of the original RCS-derived image, and the right one displays the corresponding manual annotations (clouds in yellow and aerosols in red).

TABLE I
ANNOTATION STATISTICS FOR THE DL4LIDAR DATASET

Subset	Images (%)	Aerosol		Cloud	
		Images with ≥1 instance	Total	Images with ≥1 instance	Total
Train	136 (63.8%)	136	208	105	673
Validation	34 (15.9%)	34	52	24	165
Test	43 (20.3%)	43	68	31	226
Total	213 (100%)	213	328	160	1064

3) Data Annotation: Training a segmentation model based on DL under the supervised paradigm requires expert knowledge encoded through a mask that delimits the cloud and aerosol pixels. Given the complexity and variability of atmospheric structures in lidar data, this process was manually supervised by domain experts to ensure the reliability of the ground truth used for training and validation.

To facilitate this task, we leveraged the Segment Anything Model 2 (SAM2) [41] as an interactive annotation assistant. Each atmospheric structure was delineated through a prompt-based interface, where experts provided a sequence of "positive" and "negative" clicks to adjust boundaries to the physical lidar signal. This expert-supervised approach ensured that SAM2's high-quality, class-agnostic polygons were strictly guided by expert interpretation. These masks were then assigned semantic labels: cloud or aerosol (see Fig. 3).

To assess annotation consistency and reproducibility, an inter-annotator agreement analysis was conducted on a representative subset of the dataset using independent expert annotations. The analysis showed high agreement between annotators and the final DL4Lidar masks, supporting the reliability and internal consistency of the dataset labels. In addition, all annotations underwent independent expert review and consensus-based validation.

Note that once aerosol and cloud segments are annotated in an image, the resulting mask can be reused for other images generated from the same time interval and vertical range. This enables the creation of multiple variants of the DL4Lidar dataset by modifying visualization parameters, such as the measurement channel or the maximum scale, without the need for re-annotation.

4) Dataset Overview: The DL4Lidar dataset includes 213 annotated RCS images at a resolution of 3880×2122 pixels. Each image may include annotations for two semantic classes: aerosol and cloud. Table I summarizes the annotation distri-

bution across the training, validation, and test subsets. For the aerosol class, all 213 images in the dataset contain at least one annotated instance, indicating a consistent presence of aerosol layers, and a total of 328 aerosol instances were annotated across the dataset. For the cloud class, there are 1064 cloud instances distributed across 160 out of 213 images (75.1 %).

B. Segmentation Model Design

To address the semantic segmentation of clouds and aerosols in lidar-derived images, we selected Mask R-CNN [31], a two-stage architecture widely used for instance segmentation, particularly suited to tasks requiring accurate localization and precise delineation of object boundaries [42]. Mask R-CNN extends Faster R-CNN [43] by adding a parallel branch that predicts pixel-level binary masks in addition to classification and bounding-box regression. This design is especially effective for problems where both object localization and precise delineation of boundaries are required.

We used the implementation of Mask R-CNN provided by the Detectron2 framework [44], built on PyTorch [45]. Detectron2 offers a modular and extensible architecture with interchangeable components, allowing for different backbone networks and feature-network variants, and provides a wide range of pre-trained models, such as ResNet-50 and ResNet-101 [46]. These models were originally trained on large-scale benchmarks, such as COCO [47], with additional checkpoints available for datasets like Cityscapes [48] and LVIS [49], which enables transfer learning. In this work, we leveraged COCO pre-trained weights and subsequently fine-tuned the models on the DL4Lidar dataset, which differs substantially from the natural image domains used during pretraining.

Fig. 4 illustrates the architecture of the Mask R-CNN model used in this work. The model is composed of three main components: a backbone network for feature extraction [optionally combined with a feature pyramid network (FPN)], a region proposal network (RPN) for generating candidate ROIs, and ROI heads that contain parallel branches for classification, bounding-box regression, and mask prediction. In the following, each main component is described.

1) *Backbone:* A convolutional neural network that extracts hierarchical feature maps from the input image. When coupled with an FPN [50], features at multiple scales are aggregated, which improves performance on structures of varying sizes.
2) *RPN:* This module generates candidate object proposals by scanning the feature maps for regions likely to contain objects, effectively narrowing down the areas considered for further processing.
3) *ROI Heads:* For each ROI, parallel branches predict 1) the object category; 2) bounding-box refinements; and 3) a binary segmentation mask. The mask branch is implemented as a small fully convolutional network that outputs a fixed-resolution mask for each instance. Its precision can be enhanced with refinements, such as PointRend [51], which applies a coarse-to-fine rendering strategy to sharpen mask boundaries.

From a computer vision perspective, ground-based lidar scenes often contain multiple separated structures (e.g., separated aerosol layers and intermittent cloud fragments) distributed across time-height coordinates. Therefore, Mask R-CNN is well-suited to localize and delineate such structures, while optional PointRend refinement can improve boundary fidelity in regions with sharp gradients.

A U-Net-based architecture [52] was additionally evaluated as a baseline during the experimental stage. However, the Mask R-CNN-based models consistently outperformed the U-Net alternative, especially for cloud segmentation. The region proposal strategy and instance-aware mask refinement of Mask R-CNN proved more effective for delineating fragmented and spatially heterogeneous atmospheric structures in lidar-derived images. Consequently, Mask R-CNN was retained as the primary segmentation framework in this work.

For reproducibility purposes, training was performed using the standard Detectron2 Mask R-CNN loss formulation, without additional class-weighting or imbalance-aware optimization strategies, and all experiments were executed using an NVIDIA GeForce RTX 4060.

C. Architecture Refinement

To identify the optimal semantic segmentation configuration for DL4Lidar, we systematically evaluated combinations of backbone architectures and feature-network variants within the Mask R-CNN framework. Each configuration was fine-tuned using fivefold cross-validation and assessed with per-class Intersection over Union (IoU), $F1$-score, and additionally the product of IoUs for the two classes. In particular, we considered the following backbone architectures.

1) **ResNet-50:** A moderately deep network offering a balance between performance and computational efficiency.
2) **ResNet-101:** A deeper variant with greater representational capacity, capable of capturing more complex patterns.
3) **ResNeXt-**101 32×8d [53]: Incorporates grouped convolutions to enhance learning capacity through increased cardinality, which has been shown to improve performance in various vision tasks.

We also evaluated the following feature-network variants.

1) **C4:** Uses the fourth residual block of the backbone as the sole feature source. It is straightforward and computationally efficient but less effective in multiscale scenarios.
2) **DC5:** Similar to C4, but modifies the fifth convolutional stage with dilated convolutions to maintain higher spatial resolution, providing finer feature maps without requiring an FPN.
3) **FPN:** Aggregates multiscale features from different backbone stages, providing superior performance across object sizes and achieving the best balance between accuracy and efficiency.

Additionally, we considered PointRend, a refinement applied to the mask head on top of FPN. Unlike conventional fixed-grid mask prediction, PointRend employs a coarse-to-fine point-based sampling strategy, focusing computation on uncertain pixels near boundaries. This module was included to evaluate whether explicit boundary refinement could improve

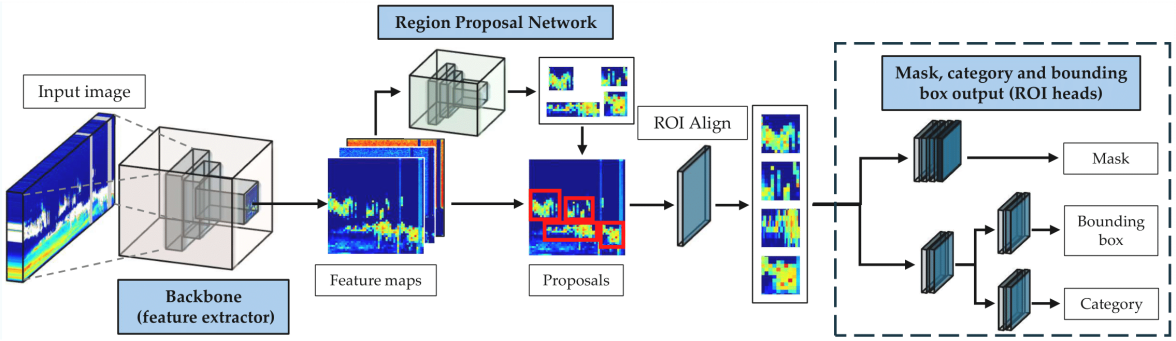


Fig. 4. Schematic architecture of the Mask R-CNN model, consisting of a backbone network with FPN, RPN, and ROI heads containing parallel branches for classification, bounding-box regression, and mask prediction. Real images from this work are used to illustrate the process.

the segmentation of atmospheric structures in lidar-derived images.

All pre-trained checkpoints were obtained from the Detectron2 model zoo. The "1×/3×" suffixes in the model names refer to different COCO training schedules used during pre-training (approximately 90k versus 270k iterations) and do not change the underlying architecture.

Given the large number of possible model configurations and hyperparameters, model selection was driven empirically, based on the mentioned standard segmentation metrics. Training was performed using fivefold cross-validation. The most effective combinations of backbone and feature-network variant were used for the remainder of experimentation and evaluation.

D. Evaluation Metrics

Since our task is semantic rather than instance segmentation, we merged all predicted instances of the same class into a single binary mask before computing semantic metrics. To assess the performance of the segmentation models, we primarily use the IoU metric, a standard measure in semantic segmentation that quantifies the overlap between predicted and reference masks. Given a predicted binary mask $\hat{M}$ and a ground truth mask M, the IoU is defined as

$$\text{IoU} = \frac{|\hat{M} \cap M|}{|\hat{M} \cup M|} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{1}$$

where TP denotes the number of true positive pixels (correctly predicted as belonging to the target class), FP the false positives (incorrectly predicted as belonging to the target class), and FN the false negatives (missed target pixels). IoU thus measures the ratio between the intersection and union of predicted and ground truth regions [54]. In addition to IoU, we also computed precision, recall, and $F1$-score, which provide complementary insights, particularly in the presence of class imbalance.

All reported metrics are computed independently for each class. However, during the model development process, a single criterion was also needed to summarize the joint performance for aerosols and clouds. Since both classes are scientifically relevant and exhibit different segmentation difficulties, this criterion should favor configurations that perform well for both classes simultaneously, rather than configurations that obtain a high score for one class while performing poorly for the other. For this reason, we additionally report the product of the class-wise IoU values: $\text{IoU Product} = \text{IoU}_{\text{cloud}} \cdot \text{IoU}_{\text{aerosol}}$.

The IoU product is therefore used as an auxiliary indicator of balanced joint performance. It is directly related to the geometric mean of the class-wise IoUs and preserves the same ranking of configurations. Compared with the arithmetic mean, this type of criterion is more sensitive to low performance in either class: if one class is poorly segmented, the product decreases substantially even if the other class achieves a high IoU. Therefore, it discourages the selection of configurations biased toward only one atmospheric component. This rationale is consistent with the use of geometric mean-based criteria in imbalanced classification problems, where balanced performance across classes is preferred over dominance by the majority or easier class [55]. In the present work, the IoU product is not intended to replace the class-wise metrics, but to complement them by providing a strict summary of whether a given configuration performs consistently for both aerosol and cloud segmentation.

III. Experimental Analysis and Results

Throughout the experimental analysis process and model development, we merged the original training and validation partitions of the DL4Lidar dataset, yielding a total of 170 images. This new set was used to perform fivefold cross-validation during the input configuration and architecture refinement phases. Once these phases ended, the original train-validation-test splits were recovered, and train and validation partitions were used to train and monitor the final model, respectively. The test subset was strictly held out throughout the entire model development process. Only after selecting the final optimal model configuration was the test set used to report the final, unbiased evaluation metrics. All training experiments were conducted using early stopping [56] and "reduce learning rate on plateau" schedule [57].

A. Selection of Data Representation and Preprocessing

The first step of the experimental process was to determine the influence of the data representation on segmentation performance when considering a multiclass problem, i.e., simultaneous segmentation of aerosol and cloud classes using

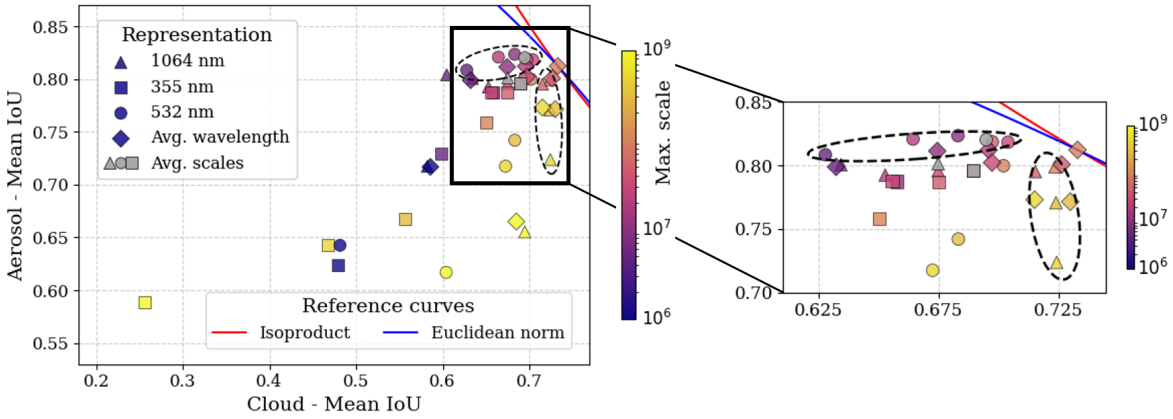


Fig. 5. Scatter plot of mean IoU values for cloud (x-axis) and aerosol (y-axis) segmentation across all data representation combinations. Marker shapes indicate the input wavelength, and color encodes the maximum scale for the RCS. The red curve traces points of equal IoU product to the maximum observed, while the blue one denotes configurations equidistant from the origin in the IoU space. Dashed ellipses group some results derived from the same wavelength channel.

the same input images, architecture, and weights. A total of 43 combinations of input data were evaluated, and they were obtained as stated in Section II-A2. To determine the optimal parameter configuration, we evaluated the performance of both single-class and multiclass predictions. Specifically, the IoU metric was computed separately for aerosol segmentation and cloud segmentation, as well as jointly for both classes by considering the product of their respective IoU scores.

To keep the number of experiments tractable, we fixed the model architecture to ResNet-101 + FPN throughout this stage. This allowed us to evaluate the impact of the data representation independently, without jointly optimizing the model configuration.

Fig. 5 summarizes the performance of the segmentation model over aerosol and cloud classes trained simultaneously. Each point in the scatter plot corresponds to a unique data representation, with the horizontal axis indicating the mean IoU for cloud detection and the vertical axis for aerosol segmentation. Marker shapes denote the input wavelength, while color encodes the maximum RCS scale bound. Two reference curves are displayed: the red curve represents points with equal IoU product to the maximum observed, and the blue line delineates equal Euclidean distance from the origin in the IoU space. The following conclusions can be drawn from the plot:

1) The highest aerosol IoUs are achieved with input data at 532 nm, particularly for scale bounds in the range 10^7–$5{\cdot}10^7$.
2) For the cloud class, the best IoUs are achieved using a wavelength of 1064 nm and larger scale bounds ($>10^8$).
3) The 355 nm wavelength consistently underperforms across both classes, indicating limited suitability for this segmentation task.
4) Representations using either excessively low or high scale bounds (shown by dark blue and bright yellow markers, respectively) yield lower performance. This degradation occurs due to image saturation (low scale) or too many RCS values falling below the saturation limit (large scale).
5) Configurations highlighted within the dashed ellipse correspond to data representations that slightly outperform others in one of the two classes. The displayed curves provide complementary perspectives on joint performance: the red hyperbola indicates the pairs of IoUs yielding the same product as the maximum obtained, while the blue circle delineates configurations that are equidistant from the origin in the IoU space.

These findings indicate that aerosol detection benefits from the use of 532 nm data, whereas cloud segmentation performs best when using 1064 nm. This behavior is consistent with the roles of both wavelengths reported in the literature: 532 nm is widely used for aerosol characterization due to its favorable balance between molecular and particle scattering [58], [59], whereas 1064 nm exhibits reduced sensitivity to aerosol backscatter and higher atmospheric transmission, which can be advantageous for identifying cloud structures, particularly under optically thick conditions [60]. The complete list of configurations and results for this first step is provided in the Appendix.

To assess that these optimal parameter configurations were not dependent on the ResNet-101 + FPN architecture used during the initial screening stage, the same data representation analysis was repeated for a representative subset of architectures: ResNet-101 + C4, ResNet-50 + FPN, and ResNet-50 + C4. In all cases, the experiments consistently identified the same wavelength-scale combinations as optimal for both aerosol and cloud segmentation, indicating that the selected data representations are robust to changes in backbone depth and feature extraction strategy.

Based on these results, a class-specific modeling strategy was adopted. Thus, two models were trained in order to segment each of the two single classes. The used experimental setup is the same as the one described previously in Section II. In this case, scale-averaged images were not considered, as the main motivation for this image processing was to enhance aerosol and cloud segmentation simultaneously. The results are reported in Table II.

This strategy led to improved results: the best IoU for the aerosol class reached 0.8425, and for the cloud class,

TABLE II

IoU for Aerosol and Cloud Models Across Data Representations. "Avg. λ" Refers to the Average RCS Image Over the Three Input Wavelengths at the Indicated Scale. The Top Three Values Within Each Panel Are in Bold

Max. scale	Aerosol IoU			
	355 nm	532 nm	1064 nm	Avg. λ
1.0e6	0.6120	0.6227	0.7316	0.7320
5.0e6	0.7224	0.8275	0.8049	0.8065
1.0e7	0.7941	0.8279	0.8203	0.8140
2.0e7	0.7964	**0.8425**	0.8180	0.8240
3.0e7	0.7857	**0.8399**	0.8225	0.8206
5.0e7	0.7791	**0.8368**	0.8190	0.8305
1.0e8	0.7751	0.8195	0.8146	0.8190
3.0e8	0.6685	0.7398	0.7983	0.7994
5.0e8	0.6558	0.7651	0.7512	0.7764
1.0e9	0.5720	0.5898	0.6716	0.7308
Max. scale	**Cloud IoU**			
	355 nm	**532 nm**	**1064 nm**	**Avg. λ**
1.0e6	0.3996	0.3762	0.5794	0.6026
5.0e6	0.6385	0.6481	0.4479	0.5285
1.0e7	0.5440	0.6918	0.3760	0.6914
2.0e7	0.6839	0.6962	0.6917	0.7328
3.0e7	0.6887	0.7146	0.6984	0.7171
5.0e7	0.6856	0.7192	0.7164	0.7267
1.0e8	0.6572	0.7191	0.7279	**0.7362**
3.0e8	0.5864	0.7085	**0.7436**	0.7345
5.0e8	0.4906	0.7038	**0.7379**	0.7314
1.0e9	0.3417	0.6311	0.7114	0.6886

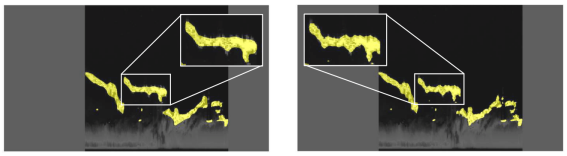
TABLE III

IoU and $F1$-Score (Mean (Std) Over Fivefold Cross-Validation) for the Top Five Architectures, Evaluated Separately on Aerosol and Cloud Classes. Best Per Metric and Class Are in Bold

Architecture	Aerosol	
	IoU	F1-score
ResNet-101 FPN (×3)	**0.850 (0.022)**	**0.913 (0.015)**
ResNeXt-101 FPN PointRend (×3)	0.847 (0.010)	0.910 (0.006)
ResNet-101 FPN PointRend (×3)	0.844 (0.020)	0.910 (0.013)
ResNet-101 DC5 (×3)	0.842 (0.012)	0.907 (0.008)
ResNet-50 DC5 (×3)	0.840 (0.010)	0.906 (0.007)
Architecture	**Cloud**	
	IoU	**F1-score**
ResNet-101 FPN (×3)	0.742 (0.015)	0.840 (0.016)
ResNeXt-101 FPN PointRend (×3)	**0.755 (0.026)**	**0.850 (0.021)**
ResNet-101 FPN PointRend (×3)	0.748 (0.020)	0.847 (0.018)
ResNet-101 DC5 (×3)	0.738 (0.016)	0.838 (0.017)
ResNet-50 DC5 (×3)	0.745 (0.019)	0.842 (0.016)

0.7436. These values correspond to relative improvements of 2.3 % and 2.6 %, respectively, compared to the baseline model trained for both classes simultaneously.

Based on these findings, the selected and optimal data representation for the aerosol segmentation task is RCS at 532 nm with a maximum scale of $2 \cdot 10^7$, while for cloud segmentation the chosen configuration is RCS at 1064 nm with a maximum scale of $3{\cdot}10^8$. These class-specific representations are used in the subsequent stage, where model architecture is refined to further enhance segmentation performance.



(a) Predictions not using PointRend. (b) Predictions using PointRend.

Fig. 6. Predictions made by two models: (a) ResNet-101 FPN without PointRend and (b) ResNet-101 FPN with PointRend. A zoomed region highlights the boundary refinement obtained when using PointRend.

B. Assessment of the Different Architectures

Having found the optimal data representation per class, we next evaluated the contribution of the network architecture to segmentation performance. We compared backbone and feature-extraction variants and the effect of PointRend as a mask-refinement module, while keeping all other training settings unchanged. Table III reports the five best-performing configurations by mean IoU and $F1$-score over fivefold cross-validation, evaluated separately for aerosols and clouds.

For aerosol segmentation, the best performance was achieved with ResNet-101 + FPN pretrained on COCO. This configuration yielded an IoU of 0.850 ± 0.022 and an $F1$-score of 0.913 ± 0.015. In this case, adding PointRend did not improve the results. This suggests that, for aerosol layers, the multiscale features extracted by FPN were already sufficient to achieve the best performance, whereas the additional boundary refinement introduced by PointRend did not provide a clear advantage.

For cloud segmentation, the highest performance was achieved using the ResNeXt-101 backbone combined with FPN and the PointRend refinement module. This architecture achieved an IoU of 0.755 ± 0.026 and an $F1$-score of 0.850 ± 0.021. These results highlight the advantages of edge-aware modules, such as PointRend, which enhance segmentation quality at fine boundaries. This improvement is illustrated in Fig. 6, where cloud edges are delineated more accurately when using PointRend.

In order to compare with our first approach, the combination of class-specific models, each using its optimal architecture, yields a product of IoUs equal to 0.6418. This represents a relative improvement of 9.3 % over the best two-class model (0.5874, see the Appendix). These results confirm that using dedicated models for each class is a superior strategy, as a single architecture trained simultaneously on both classes does not reach such a level of performance.

Interestingly, the top-performing models for each class also perform competitively on the other class, suggesting a certain degree of generality. However, class-specific optimization remains the most effective approach for maximizing absolute performance. Detailed metrics for all evaluated architectures are provided in the Appendix.

C. Selection of the Best Model

1) Overall Performance: To build the final two-model segmentation system, the following procedure was implemented.

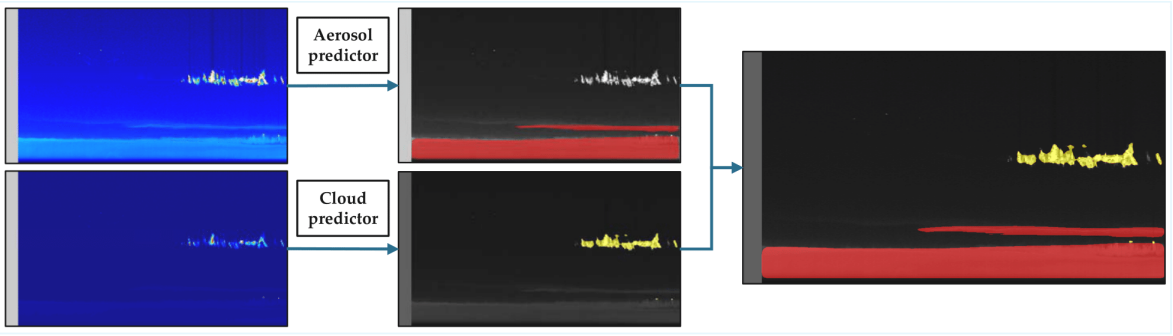


Fig. 7. Segmentation obtained by applying both predictors, aerosol (above) and cloud (below), to one of the images in the dataset.

TABLE IV

MEAN (STD) OF IOU, $F1$-SCORE, AND PIXEL ACCURACY OVER FIVEFOLD CROSS-VALIDATION FOR AEROSOL AND CLOUD CLASSES BY DATASET SUBSET

Subset	IoU		F1-score		Pixel accuracy
	Aerosol	Cloud	Aerosol	Cloud	Overall
Train	0.816 (0.150)	0.791 (0.115)	0.890 (0.107)	0.878 (0.080)	0.973 (0.035)
Validation	0.820 (0.139)	0.745 (0.170)	0.895 (0.091)	0.839 (0.164)	0.976 (0.030)
Test	0.814 (0.143)	0.753 (0.089)	0.890 (0.100)	0.856 (0.063)	0.975 (0.029)

First, the RCS data were selected according to the optimal configuration for each task, that is, using the wavelength and maximum scale bound that yielded the best performance for aerosols and clouds, respectively. The RCS values were then plotted as images, which served as input to the trained models. The outputs of both models were subsequently combined, allowing the predictions to be associated with the same underlying RCS data. Fig. 7 illustrates this process and shows the resulting predictions, with aerosol detections shown in red and cloud detections in yellow.

Table IV reports the quantitative performance of the segmentation system across the aerosol and cloud classes, evaluated separately on the training, validation, and test subsets. All values are reported as mean (standard deviation) across each subset.

On the one hand, the model achieves an IoU of 0.814 on the test set for aerosols, closely matching the results on the training (0.816) and validation (0.820) subsets, with standard deviations in the range of 0.14–0.15. The corresponding $F1$-scores are also stable, ranging from 0.890 to 0.895. These results demonstrate the model's ability to reliably detect aerosol presence across varying scenes, without evidence of overfitting.

Aerosol segmentation is generally more challenging due to less distinct visual patterns and ambiguous spatial boundaries compared to cloud structures. The observed consistency across all subsets indicates that the model has effectively learned to generalize from subtle patterns, likely by leveraging spatial context and feature hierarchies. This suggests robustness to the significant variability in aerosol morphology and distribution present in the dataset.

On the other hand, the model exhibits strong generalization capabilities for cloud segmentation, achieving an IoU of 0.753 and an $F1$-score of 0.856 on the test set. These values are comparable to those obtained on the training (IoU = 0.791, $F1$ = 0.878) and validation (IoU = 0.745, $F1$ = 0.839) subsets, indicating stable performance across data splits.

Notably, the standard deviation of the cloud IoU on the validation set is higher ($\pm$0.170) compared to the training and test subsets, suggesting greater heterogeneity in cloud appearance and structure within that partition. Factors, such as multilayered clouds, partial occlusions, and thin cirrus formations, likely contribute to this variability and increased segmentation uncertainty. Despite these challenges, the model maintains robust performance, indicating that it effectively captures cloud-related features under diverse atmospheric conditions.

Aerosol segmentation displays greater consistency across data subsets in both IoU and $F1$-score. However, cloud segmentation shows slightly higher mean values in the training and test sets. This difference likely reflects the distinct spatial characteristics of each class: cloud structures tend to be more compact and well-defined, whereas aerosol layers are often diffuse, horizontally extended, and more difficult to delineate precisely, posing different segmentation challenges.

The narrow performance gap between the training and test sets across all metrics further supports the effectiveness of the training strategy as well as the model segmentation capabilities and robustness. Overall, the metrics show consistently high performance and generalization capacity after training.

2) Detailed Performance According to Sizes: To gain further insight into the model's behavior, we analyzed its performance across different sizes of atmospheric elements. Aerosol and cloud annotations were grouped in quartiles according to the total number of pixels per image of the corresponding class in the ground truth. This allowed us to explore how segmentation performance varies with the different sizes of aerosol masks and the spatial distribution of clouds across the images.

Table V summarizes the segmentation performance for the aerosol class across the four quartiles, and Fig. 8 shows examples of images assigned to each of the quartiles. The model shows consistently high performance for small and

TABLE V

SEGMENTATION PERFORMANCE FOR THE AEROSOL CLASS, STRATIFIED BY QUARTILES OF GROUND-TRUTH MASK SIZE (Q1: SMALLEST, Q4: LARGEST). ALL METRICS ARE COMPUTED ON THE TEST SET AND REPORTED AS MEAN (STD). THE MEAN AEROSOL LAYER HEIGHT FOR EACH QUARTILE IS REPORTED IN THE SECOND COLUMN

Quartile	Mean height (m)	IoU	F1-score	Recall	Precision
Q1	782	0.820 (0.105)	0.897 (0.069)	0.930 (0.029)	0.873 (0.111)
Q2	1454	0.833 (0.201)	0.894 (0.149)	0.860 (0.201)	0.951 (0.054)
Q3	1966	0.820 (0.142)	0.894 (0.096)	0.936 (0.057)	0.874 (0.155)
Q4	3435	0.786 (0.122)	0.875 (0.080)	0.891 (0.087)	0.880 (0.140)

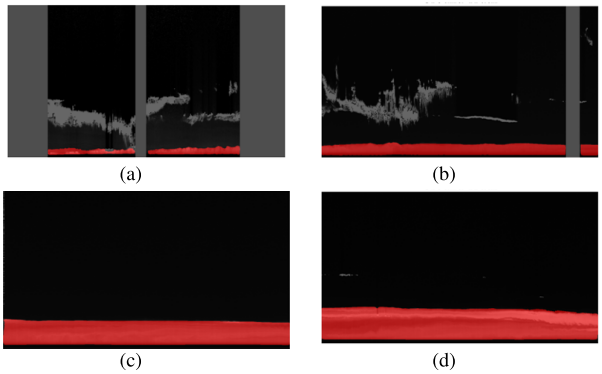


Fig. 8. Dataset entries belonging to each quartile for aerosol class, sorted from top-left to bottom-right according to increasing size: (a) Q1, (b) Q2, (c) Q3, and (d) Q4.

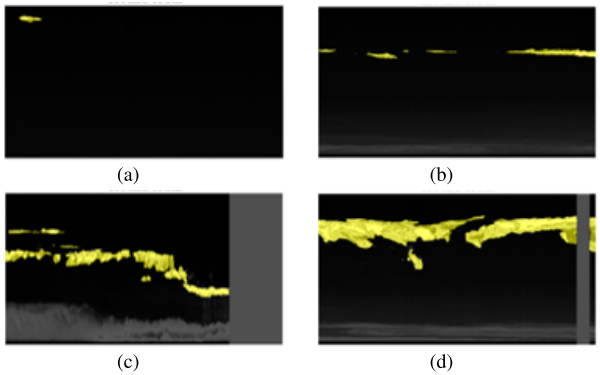


Fig. 9. Dataset entries belonging to each quartile for cloud class, sorted from top-left to bottom-right according to increasing amount of pixels: (a) Q1, (b) Q2, (c) Q3, and (d) Q4.

medium-sized aerosol layers, with mean IoU values ranging from 0.820 to 0.833 in Q1-Q3. A slight performance decrease is observed in Q4 (IoU = 0.786), which also exhibits increased variance, possibly reflecting the morphological diversity of the largest structures. However, all quartiles retain high $F1$-scores, confirming that the model can reliably segment aerosol layers across a wide range of spatial extents.

To complement this analysis, we also examined whether the model's performance varied with the vertical thickness of the aerosol layers, computed as the average number of ground truth pixels per image column. While the primary grouping by mask size already captures spatial extent in two dimensions, this thickness-based analysis provides a modest refinement of the previous grouping, accounting for gaps in the measured RCS. This analysis is shown in Table VI, where the metrics for each average aerosol thickness are reported. The segmentation performance also reflects the model's ability to adapt to the vertical structure of the aerosol layers. The best IoU scores are observed for moderately thick aerosol columns, between approximately 1000 and 2000 m (Q2–Q3), with mean values exceeding 0.84. These findings confirm the model's suitability for detecting typical aerosol configurations observed in the dataset. For thinner cases below 1 km (Q1), although the IoU slightly decreases, the high recall (0.87) and $F1$-score (0.86) indicate that the model is still effective at identifying these subtle structures, even if their precise contours are harder to delineate. In the case of the thickest aerosol columns, above 2 km (Q4), the IoU drops more noticeably, yet the model maintains a strong recall (0.90) and competitive $F1$-score (0.85), suggesting that it successfully captures most of the target structure despite increased morphological complexity. These findings reinforce the model's robustness across a broad range of vertical aerosol extents.

A similar analysis was conducted for the cloud class, using the same quartile-based grouping by ground truth mask size. Table VII presents the results corresponding to each group, and Fig. 9 shows an example of images belonging to each group. The model delivers consistent performance across the full range of cloud image groups, with mean IoU values exceeding 0.72 in all quartiles. The highest IoU is achieved for the images with the highest amount of cloud pixels in their ground truth annotations (Q4, 0.775), closely followed by the one with the least amount of pixels (Q1, 0.769), which may reflect the well-defined spatial signatures typically associated with isolated or compact cloud structures.

In terms of overall segmentation quality, macro-averaged $F1$-scores remain above 0.83 across all groups, confirming the model's capacity to accurately identify cloud regions regardless of their extent. The recall and precision values exhibit a balanced profile in Q1 and Q4, while Q2 and Q3 display slightly more variability. Notably, the performance variability in Q4 is lower than in earlier quartiles, suggesting more stable predictions for extended cloud regions. These results highlight the model's generalization ability in capturing cloud structures under diverse spatial configurations and scales.

D. Evaluation of Generalization on a Different Ground-Based Lidar Instrument

To assess the generalization capabilities of the models trained on the ALHAMBRA lidar dataset, we applied them directly, without any retraining or fine-tuning, to an

TABLE VI

SEGMENTATION METRICS FOR THE AEROSOL CLASS, GROUPED BY AVERAGE VERTICAL THICKNESS (M) OF THE AEROSOL LAYER IN THE TEST SET. VALUES ARE PRESENTED AS MEAN (STD)

Quartile	Thickness range (m)	IoU	F1-score	Recall	Precision
Q1	< 1040	0.7752 (0.1739)	0.8617 (0.1315)	0.8735 (0.1650)	0.8713 (0.1125)
Q2	1040–1359	0.8809 (0.1263)	0.9315 (0.0849)	0.9227 (0.1265)	0.9502 (0.0455)
Q3	1359–2000	0.8464 (0.0805)	0.9149 (0.0484)	0.9165 (0.0562)	0.9194 (0.0818)
Q4	> 2000	0.7580 (0.1506)	0.8544 (0.1027)	0.9026 (0.0899)	0.8405 (0.1796)

TABLE VII

SEGMENTATION PERFORMANCE FOR THE CLOUD CLASS, STRATIFIED BY QUARTILES OF GROUND-TRUTH MASK SIZE (Q1: SMALLEST, Q4: LARGEST). ALL METRICS ARE COMPUTED ON THE TEST SET AND REPORTED AS MEAN (STD). THE SECOND COLUMN INDICATES THE RANGE OF CLOUD MASK SIZES AS A PERCENTAGE OF THE TOTAL IMAGE AREA (3880×2122 PX)

Quartile	Mask size [%]	IoU	F1-score	Recall	Precision
Q1	< 1.38	0.769 (0.132)	0.863 (0.097)	0.853 (0.156)	0.892 (0.056)
Q2	1.38–3.72	0.742 (0.079)	0.850 (0.052)	0.883 (0.092)	0.842 (0.136)
Q3	3.72–5.87	0.721 (0.082)	0.836 (0.060)	0.815 (0.118)	0.873 (0.064)
Q4	> 5.89	0.775 (0.052)	0.873 (0.033)	0.886 (0.058)	0.868 (0.085)

TABLE VIII

KEY ALHAMBRA, MULHACEN, AND ATLID LIDAR INSTRUMENTAL CHARACTERISTICS AND DATA USED

	ALHAMBRA	MULHACEN	ATLID
Platform	Ground-based	Ground-based	Spaceborne
Temp. resolution	60 s	60 s	~ 0.143 s
Vert. resolution	3.75 m	7.5 m	103 m
Channels (nm)	355, 532, 1064	355, 532, 1064	355
Data type	RCS	RCS	Attenuated backscatter

independent dataset acquired using the MULHACEN lidar system. MULHACEN is another multiwavelength Raman lidar featuring the same emission channels as ALHAMBRA, which was operated from 2005 to 2023. It is also located in Granada, belonging to the AGORA infrastructure and also to EARLINET/ACTRIS. This evaluation serves to test the robustness of our DL-based approach under instrumental variability, and to evaluate how well the trained models can maintain their performance when applied to data from a different lidar system, in a zero-shot cross-instrument scenario. The key technical specifications of MULHACEN, compared to ALHAMBRA, are summarized in Table VIII.

The evaluation on MULHACEN was performed using a subset of RCS measurements collected between January and August 2020, comprising 46 annotated images. Each image contained at least one aerosol instance, and 29 of them featured both aerosol and cloud structures. Preprocessing was conducted identically to that of ALHAMBRA (refer to Section II) to ensure compatibility with the model's input pipeline. Moreover, the model weights and architecture were left unchanged. Ground truth annotations were independently produced using the same procedure described in Section II-A. Due to the aging of MULHACEN's laser, a slightly lower maximum scale ($1.0 \cdot 10^7$ instead of $2.0 \cdot 10^7$) was used for the aerosol images representation to account for the weaker received signal [61].

Table IX summarizes the segmentation performance of the trained models on the ALHAMBRA and MULHACEN test datasets. ALHAMBRA metrics are shown as mean values only (refer to Table IV for full details).

As shown in Table IX, the model trained exclusively on ALHAMBRA data generalizes well to the MULHACEN dataset. For the aerosol class, the IoU drops only slightly from 0.814 to 0.801 (a relative decrease of $\sim$1.6%), and the $F1$-score decreases marginally from 0.890 to 0.883 (-0.8%). These minimal differences confirm the strong robustness of the model to domain shifts in aerosol segmentation, with comparable average performance and stable variability across instruments.

For cloud segmentation, generalization is equally successful. In fact, both the IoU and $F1$-score slightly improve when applied to the MULHACEN data, rising from 0.753 to 0.774 in IoU, and from 0.856 to 0.887 in $F1$-score. This result is particularly noteworthy given the potential challenges associated with hardware degradation and instrument-specific signal characteristics. The high precision (0.90) and recall (0.85) values further support the model's ability to capture cloud structures effectively, even in the presence of potential signal attenuation or noise.

It is worth noting that although the average performance improves on MULHACEN, the associated standard deviation is also higher, indicating greater variability across samples. This suggests that the model does not systematically outperform its performance on ALHAMBRA, but remains robust under different acquisition conditions.

Pixel accuracy shows a mild drop from 0.975 to 0.947, which is expected given the domain shift. However, since this metric is dominated by the background class, it provides limited interpretive value in isolation and should be considered with caution.

Overall, these results demonstrate the strong generalization capacity of the model across instruments. The aerosol class maintains performance with minimal degradation, while cloud segmentation not only remains consistent but even slightly improves. This supports the viability of deploying models trained on a single, well-annotated system (ALHAMBRA) to similar lidar systems (e.g., MULHACEN) without requiring retraining or adaptation.

TABLE IX
COMPARISON OF MODEL PERFORMANCE ON ALHAMBRA AND MULHACEN TEST DATA. METRICS ARE REPORTED AS MEAN (STD)

Lidar	IoU		F1-score		Pixel accuracy
	Aerosol	Cloud	Aerosol	Cloud	Overall
ALHAMBRA	0.814	0.753	0.890	0.856	0.975
MULHACEN	0.801 (0.152)	0.774 (0.155)	0.883 (0.117)	0.887 (0.040)	0.947 (0.050)

E. Evaluation of Generalization on Satellite Lidar Observations

To evaluate the generalization capacity of the proposed approach beyond ground-based observations, the cloud segmentation model was applied to satellite lidar data from the ATLID onboard the EarthCARE mission [15]. The selected variable corresponds to the total attenuated backscatter coefficient at 355 nm from the Level 2A A-EBD product from baseline BA, which is physically proportional to the RCS used in our ground-based experiments and represents the most comparable observable available from ATLID. The main differences between ATLID and the ground-based lidar systems used in this study are summarized in Table VIII. Minor preprocessing adjustments were required to adapt the satellite input images to the model configuration, such as image vertical inversion to account for the nadir-looking geometry from space. Additionally, shorter time periods were used to mitigate the effect of the satellite's orbital velocity. Preliminary tests demonstrated that the model can successfully reproduce cloud structures from spaceborne measurements.

In particular, we have selected three images with cloud presence, corresponding to frame B of orbits 6677 and 6706, and frame D of orbit 6699. The cloud segmentation model was applied to them, and the corresponding cloud masks were inferred. The results are satisfactory, although minor artifacts might appear near lower cloud boundaries, typically within highly attenuated regions of the signal. If we derive the cloud top height (CTH) from the masks as the top-most cloud pixel in each timestamp, the mentioned artifact is not relevant, and it allows us to derive the temporal evolution of the CTH. As ATLID provides a CTH product as well, both time series are plotted in Fig. 10, using red for the ATLID product and blue for our DL CTH retrieval, so that they can be compared. The reference product exhibits discretized height levels due to its binning scheme, whereas the proposed method yields a continuous representation of the upper cloud boundary, capturing subtle vertical variations in the cloud top morphology. Preliminary visual assessment across these orbits indicates a strong correspondence between both CTH estimates, suggesting that the model captures the vertical structures observed by ATLID. The EarthCARE product tends to underestimate the CTH and captures some regions as false CTH, while the segmentation approach might miss some regions where clouds are very thin [e.g., Fig. 10(c)].

These findings demonstrate that a model trained exclusively on ground-based lidar data can successfully reproduce the main cloud structures detected by a satellite-borne lidar operating with a different configuration. This ability to generalize across instruments indicates that the model internalizes robust, instrument-independent morphological features of clouds rather than sensor-specific radiometric cues. To our knowledge, this represents one of the first demonstrations of cross-platform generalization of DL-based cloud segmentation models from ground-based to spaceborne lidar observations.

This preliminary implementation focused on clouds, as aerosol layers remain more challenging to detect from spaceborne lidars due to strong signal attenuation near the surface and lower signal-to-noise ratios [62], [63]. A dedicated investigation on aerosol detection and wavelength adaptation will be addressed in future work. It should also be noted that the ATLID backscatter data used here are currently flagged as "Quality Status: DEGRADED," reflecting possible refinement in future-reviewed products. Consequently, this analysis should be interpreted as a preliminary demonstration of feasibility. Once nominal-quality products become available, a comprehensive evaluation of the model's transferability to satellite lidar data will be undertaken. Nonetheless, the present results already show that a DL model trained with ground-based lidar images can be effectively extrapolated to spaceborne observations, opening a pathway toward unified classification frameworks applicable across ground-based and spaceborne lidars.

IV. DISCUSSION

The results presented in this work demonstrate that DL-based semantic segmentation provides an effective and physically consistent framework for identifying aerosols and clouds directly from ground-based lidar RCS. The systematic evaluation of wavelengths, visualization scales, and architectural variants shows that segmentation performance is strongly conditioned by the interaction between the physical characteristics of the lidar signal and the chosen data representation. The class-dependent optimal inputs, 532 nm for aerosols and 1064 nm for clouds, reflect the distinct ways in which each atmospheric structure manifests in the elastic backscatter signal, illustrating the importance of coupling physical understanding with data-driven model design.

Architecture selection further reveals that the optimal configuration is not universal but depends on the spatial morphology of the atmospheric structures. Aerosol layers, generally diffuse and vertically smooth, benefited from the representational capacity of a ResNet-101 + FPN backbone, whereas cloud structures, characterized by sharper edges and well-defined boundaries, exhibited improved delineation when PointRend-based mask refinement was applied. These results highlight that model complexity should be aligned with the

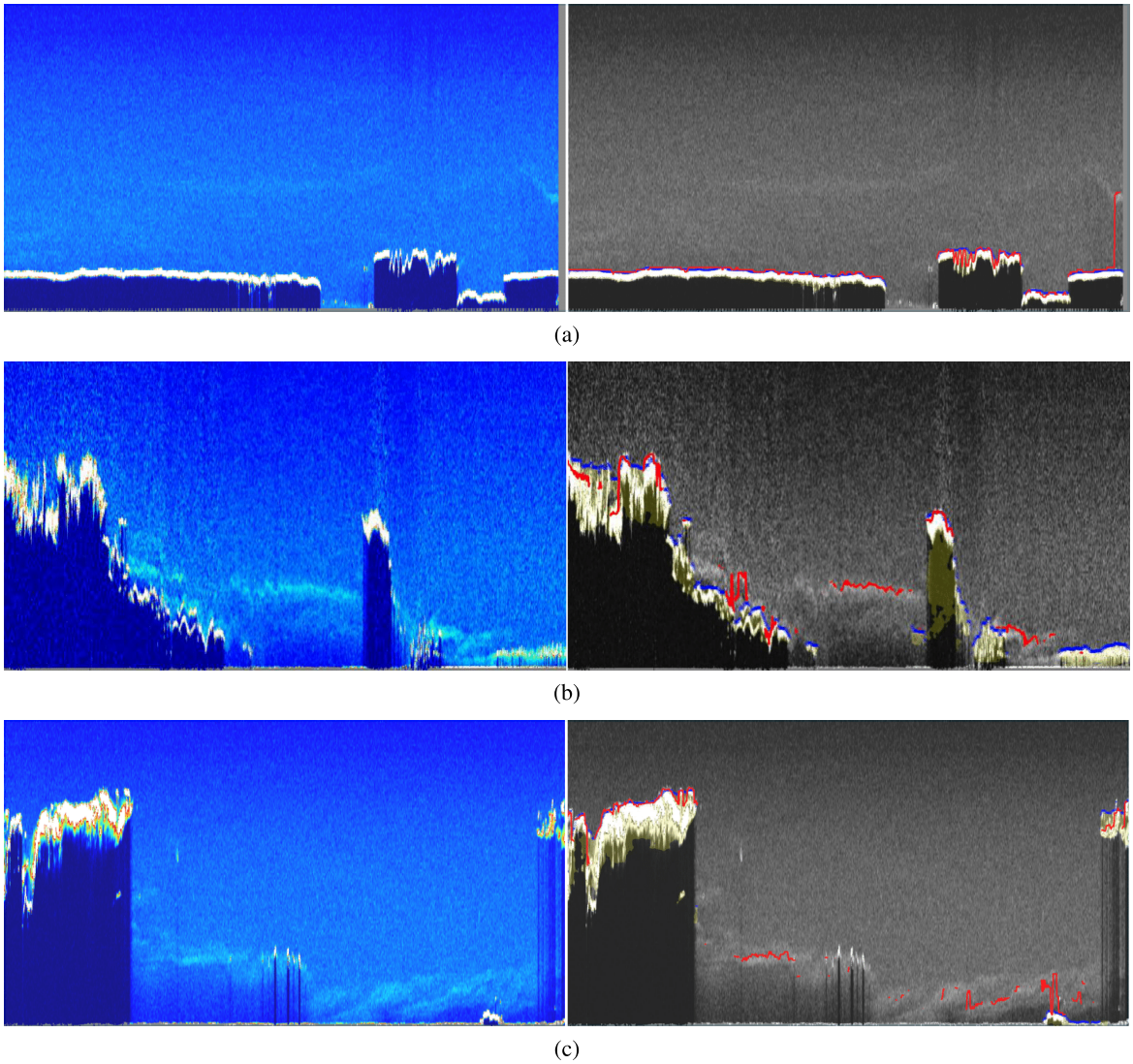


Fig. 10. Input images of the cloud segmentation model on the left, derived from ATLID total attenuated backscatter at 355 nm at 10^{-5} $m^{-1}sr^{-1}$ from L2A A-EBD baseline BA. On the right, predicted cloud masks in yellow, ATLID CTH product in red, and obtained CTH from the masks in blue. (a) Orbit 6677B, on 2025-08-01 from 01:49:57 to 01:53:00. (b) Orbit 6699D, on 2025-08-02 from 12:05:43 to 12:08:46. (c) Orbit 6706B, on 2025-08-02 from 22:30:41 to 22:33:44.

physical and geometric properties of the target class rather than treated uniformly across tasks.

The models also demonstrated strong transferability across lidar instruments. When applied directly to the independent MULHACEN lidar system, segmentation performance was preserved for aerosols and slightly improved for clouds despite differences in laser power, detector response, and signal degradation. This robustness suggests that the learned features primarily encode atmospheric morphology rather than instrument-specific radiometric signatures, an essential requirement for deployment across lidar networks.

A preliminary transfer experiment using ATLID observations from the EarthCARE mission further supports this interpretation. Despite differences in wavelength, geometry, and sampling characteristics, the ground-trained cloud model reproduced the main cloud structures and generated CTH estimates consistent with the reference product. These findings indicate that the learned representations capture instrument-independent structural patterns, opening paths for unified classification frameworks applicable to both ground-based and satellite lidars.

The stratified performance analysis by aerosol layer size and cloud coverage provides additional insight into the model's operational reliability. Aerosol segmentation remained stable across a wide range of vertical extents, with only moderate degradation for thick layers. Cloud segmentation similarly demonstrated consistent performance across quartiles, with reduced variability for spatially extensive cloud fields. These results indicate that the proposed segmentation system is robust across diverse atmospheric scenarios.

Finally, the development of an operationally oriented inference pipeline capable of processing new RCS images in near real time underscores the practical relevance of this work. The entire workflow, from dataset construction to architecture optimization and deployment, is reproducible and extensible, providing a foundation for future developments that incorporate additional lidar observables or hybrid physical-ML retrieval schemes.

TABLE X

MEAN IOU FOR THE COMBINED MODEL OF AEROSOL AND CLOUD CLASSES ACROSS DATA REPRESENTATIONS. "AVG. λ" COMBINES THE RCS AT 355, 532, AND 1064 NM AT THE INDICATED SCALE. "AVG. SCALES" ARE COMPUTED AT A FIXED WAVELENGTH USING THE SCALES MARKED WITH $*$ IN THAT COLUMN. COLUMN-WISE MAXIMUM VALUES ARE IN BOLD FOR EACH INDIVIDUAL IOU TABLE

Max. scale	IoU — Aerosol				IoU — Cloud				IoU aerosol × IoU cloud			
	355 nm	532 nm	1064 nm	Avg. λ	355 nm	532 nm	1064 nm	Avg. λ	355 nm	532 nm	1064 nm	Avg. λ
1.0e6	0.6241	0.6425	0.7174	0.7168	0.4795	0.4815	0.5824	0.5859	0.2993	0.3094	0.4178	0.4200
5.0e6	0.7296	0.8084	**0.8039***	0.7988	0.5984	0.6279	0.6047*	0.6323	0.4366	0.5076	0.4861	0.5051
1.0e7	0.7874	**0.8234***	0.8005	0.8113	0.6578	0.6831*	0.6343	0.6745	0.5180	0.5625	0.5078	0.5472
2.0e7	**0.7959***	0.8207	0.7924	**0.8122**	**0.6895***	0.6644	0.6528	0.6960	0.5488	0.5453	0.5173	0.5653
3.0e7	0.7875	0.8182*	0.7958	0.8022	0.6558	**0.7040***	0.6749	0.6974	0.5164	0.5760	0.5371	0.5595
5.0e7	0.7869	0.8185	0.7951	0.8007	0.6751	0.6975	0.7154	0.7264	0.5312	0.5709	0.5688	0.5816
1.0e8	0.7584	0.7997	0.7990*	0.8022	0.6504	0.7022	0.7236*	**0.7322**	0.4933	0.5615	0.5782	0.5874
3.0e8	0.6677	0.7420	0.7706	0.7714	0.5565	0.6833	0.7240	0.7300	0.3716	0.5070	0.5579	0.5631
5.0e8	0.6432	0.7174	0.7237*	0.6948	0.4674	0.6726	**0.7243***	0.7151	0.3006	0.4825	0.5242	0.5528
1.0e9	0.5884	0.6170	0.6553	0.6651	0.2559	0.6041	0.6949	0.6851	0.1506	0.3727	0.4554	0.4557
Avg. scales	0.7959	**0.8200**	0.8019		0.6678	**0.6948**	0.6747		0.5488	0.5699	0.5410	

TABLE XI

PERFORMANCE METRICS, MEAN (STD), FOR AEROSOL SEGMENTATION ACROSS ALL TESTED ARCHITECTURES. RESULTS ARE AVERAGED OVER FIVE VALIDATION FOLDS. THE LAST COLUMN REPORTS THE PRODUCT OF THE FOUR METRICS

Architecture	IoU	Precision	Recall	F1-score	Product
ResNet-101 FPN (×3)	0.850 (0.022)	0.919 (0.025)	0.920 (0.015)	0.913 (0.015)	0.656
ResNeXt-101 FPN PointRend (×3)	0.847 (0.010)	0.921 (0.008)	0.912 (0.014)	0.910 (0.006)	0.647
ResNet-101 FPN PointRend (×3)	0.844 (0.020)	0.934 (0.014)	0.897 (0.026)	0.910 (0.013)	0.642
ResNet-101 DC5 (×3)	0.842 (0.012)	0.900 (0.017)	0.929 (0.016)	0.907 (0.008)	0.639
ResNet-50 DC5 (×3)	0.840 (0.010)	0.906 (0.016)	0.922 (0.010)	0.906 (0.007)	0.635
ResNet-50 C4 (×3)	0.837 (0.022)	0.901 (0.015)	0.925 (0.012)	0.905 (0.016)	0.631
ResNet-101 C4 (×3)	0.838 (0.019)	0.901 (0.016)	0.925 (0.007)	0.904 (0.013)	0.631
ResNet-50 FPN (×3)	0.836 (0.011)	0.905 (0.017)	0.918 (0.014)	0.903 (0.007)	0.627
ResNet-50 C4 (×1)	0.835 (0.015)	0.896 (0.014)	0.928 (0.012)	0.902 (0.011)	0.626
ResNet-50 FPN PointRend (×3)	0.834 (0.017)	0.924 (0.009)	0.895 (0.019)	0.903 (0.012)	0.623
ResNet-50 FPN PointRend (×1)	0.834 (0.009)	0.921 (0.011)	0.898 (0.016)	0.902 (0.006)	0.622
ResNet-50 DC5 (×1)	0.830 (0.018)	0.888 (0.017)	0.926 (0.010)	0.898 (0.011)	0.613
ResNet-50 FPN (×1)	0.827 (0.013)	0.901 (0.021)	0.913 (0.010)	0.898 (0.008)	0.611

TABLE XII

PERFORMANCE METRICS, MEAN (STD), FOR CLOUD SEGMENTATION ACROSS ALL TESTED ARCHITECTURES. RESULTS ARE AVERAGED OVER FIVE VALIDATION FOLDS. THE LAST COLUMN SHOWS THE PRODUCT OF THE FOUR METRICS

Architecture	IoU	Precision	Recall	F1-score	Product
ResNeXt-101 FPN PointRend (×3)	0.754 (0.025)	0.895 (0.023)	0.835 (0.041)	0.850 (0.021)	0.479
ResNet-101 FPN PointRend (×3)	0.748 (0.020)	0.894 (0.018)	0.827 (0.026)	0.847 (0.018)	0.468
ResNet-50 DC5 (×3)	0.745 (0.019)	0.854 (0.009)	0.860 (0.026)	0.842 (0.016)	0.461
ResNet-101 FPN (×3)	0.742 (0.015)	0.868 (0.012)	0.838 (0.021)	0.840 (0.016)	0.454
ResNet-101 DC5 (×3)	0.738 (0.016)	0.852 (0.015)	0.855 (0.028)	0.838 (0.017)	0.451
ResNet-50 DC5 (×1)	0.737 (0.020)	0.836 (0.012)	0.865 (0.028)	0.834 (0.017)	0.445
ResNet-50 FPN (×3)	0.734 (0.025)	0.868 (0.022)	0.829 (0.031)	0.834 (0.022)	0.441
ResNet-50 FPN (×1)	0.735 (0.024)	0.855 (0.033)	0.840 (0.024)	0.834 (0.020)	0.440
ResNet-50 FPN PointRend (×3)	0.730 (0.010)	0.890 (0.023)	0.805 (0.019)	0.834 (0.013)	0.436
ResNet-101 C4 (×3)	0.728 (0.017)	0.843 (0.011)	0.849 (0.029)	0.832 (0.014)	0.433
ResNet-50 FPN PointRend (×1)	0.729 (0.027)	0.872 (0.011)	0.819 (0.036)	0.828 (0.022)	0.431
ResNet-50 C4 (×3)	0.725 (0.018)	0.826 (0.012)	0.868 (0.032)	0.829 (0.015)	0.431
ResNeXt-101 FPN (×3)	0.728 (0.018)	0.863 (0.009)	0.821 (0.034)	0.825 (0.020)	0.425
ResNet-50 C4 (×1)	0.720 (0.021)	0.820 (0.012)	0.861 (0.032)	0.825 (0.019)	0.419

Beyond the direct generalization experiments presented here, the trained model may also provide a useful baseline for the community, allowing for cross-instrument lidar segmentation studies. In cases where no annotated data are available for a new lidar system, the model can be applied directly in a training-free transfer setting, as demonstrated in this work. If a limited number of annotated samples is available, partial fine-tuning of the prediction heads or selected high-level layers may provide an efficient adaptation strategy with reduced labeling effort. For larger target-domain datasets, full fine-tuning or mixed-source training combining ALHAMBRA and target-domain samples could further improve migration performance. Therefore, the release of the trained model and associated resources is intended to facilitate both direct application and further adaptation by the lidar and remote sensing community.

V. Conclusion

This study demonstrates that DL-based semantic segmentation can accurately and robustly identify aerosol and cloud structures directly from lidar RCS images. By constructing an expert-annotated dataset, DL4Lidar, optimizing wavelength and scaling representations grounded in lidar physics, and selecting architectures tailored to atmospheric morphology, the proposed framework achieves high segmentation accuracy and strong cross-instrument generalization. The models generalize effectively to an independent ground-based lidar and exhibit promising performance when applied to EarthCARE ATLID observations, suggesting that lidar image morphology can be learned in a sensor-independent manner. These results highlight the potential of computer vision to advance automated atmospheric profiling and motivate future work integrating additional lidar observables, hybrid physical-ML approaches, and unified methodologies for both ground-based and satellite lidar systems.

Appendix

This appendix provides the complete experimental results supporting the analyses presented in Section III. Table X reports the performance of the combined aerosol-cloud segmentation model for the different data representations considered. Tables XI and XII summarize the complete performance metrics obtained for all evaluated architectures in the aerosol and cloud segmentation experiments, respectively.

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